

Chelsea Brackett

CSCI E-102 Econometrics and Causal Inference with R
Part I of Assignment 10

Consider the following Tobit model:

$$Y_i = \begin{cases} Y_i^*, & \text{if } Y_i^* \geq 0, \\ 0, & \text{if } Y_i^* < 0, \end{cases}$$

where

$$Y_i^* = \beta_0 + \beta_1 X_i + \varepsilon_i$$

is a *latent* (unobserved) variable, and the error is assumed to follow Normal distribution:

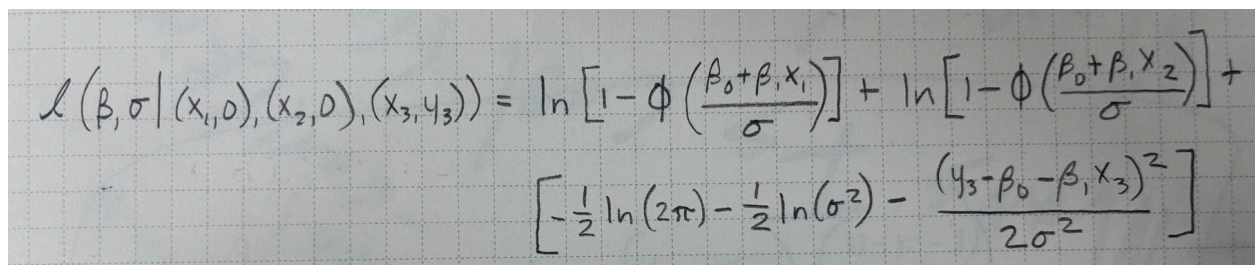
$$\varepsilon_i \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma^2), \quad i \in \{1, 2, \dots, n\}.$$

If we observe $(x_1, 0), (x_2, 0), (x_3, y_3)$, where $y_3 > 0$, what is the log-likelihood function

$$\ell(\beta_0, \beta_1, \sigma \mid \underbrace{(x_1, 0), (x_2, 0), (x_3, y_3)}_{\text{data}})$$

for this model? Please make sure that the log-likelihood function depends on $\beta_0, \beta_1, \sigma, x_1, x_2, x_3$, and y_3 only.

SOLUTION:


$$\begin{aligned} \mathcal{L}(\beta, \sigma \mid (x_1, 0), (x_2, 0), (x_3, y_3)) &= \ln \left[1 - \Phi \left(\frac{\beta_0 + \beta_1 x_1}{\sigma} \right) \right] + \ln \left[1 - \Phi \left(\frac{\beta_0 + \beta_1 x_2}{\sigma} \right) \right] + \\ &\quad \left[-\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{(y_3 - \beta_0 - \beta_1 x_3)^2}{2\sigma^2} \right] \end{aligned}$$